

# **Ch 5: Elasticity and Its Application**

How much quantity responds, and why it pays to know

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

# From direction to size

## Chapter 4 answered

Which **direction** does the price move?

Which way does quantity go?

Supply falls  $\Rightarrow$  price up, quantity down. Every time.

## Chapter 5 asks

By **how much**?

## Why the size decides everything

A cigarette tax: mostly less smoking, or mostly more revenue?

A record harvest: good or bad news for farmers?

You raise your price 25%: do you earn more or less?

Same diagram, opposite answers, depending on **one number**

Monday: the Kerala paper, then measuring demand · Wednesday: revenue, supply, applications

# The reading: Kerala, 1997-2001

## The market

Sardine fishermen on the Kerala coast · fifteen beach markets

Markets open 5:00-8:00 AM · one market a day per boat · no storage, no ice

Over 70% of adults eat fish daily

## The problem

At sea you cannot see any price.  
You pick a beach before you know what it pays.

## One morning: Tuesday, Jan 14, 1997

| Market   | Price (Rs/kg) |
|----------|---------------|
| Badagara | 0             |
| Chombala | 9.9           |
| Quilandi | 9.8           |

All within 15 km. Jensen (2007), Table I.

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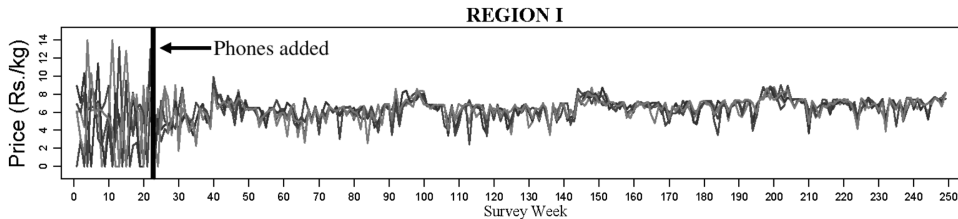
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Eleven boats at Badagara dumped the catch in the sea while buyers 15 km away went home empty — as much as 3,400 Rs in profit forgone per boat

# The graph everyone remembers

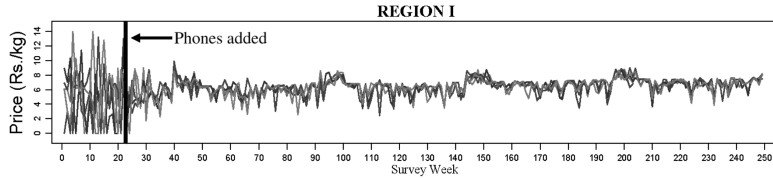


One line per beach market, Region I. 7:30–8:00 AM sardine price, 2001 Rs/kg. Jensen (2007), Fig. IV.

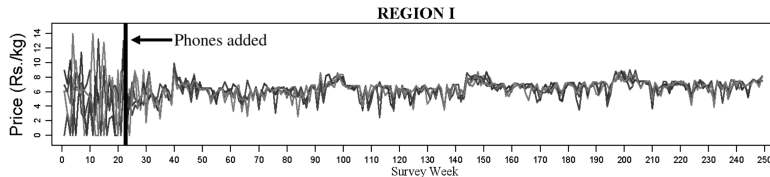
## Question

- (a) One morning, one market pays 14 Rs/kg and another pays 0. How can fish be worth zero?
- (b) After week 23 the lines sit on top of each other. Who gained?

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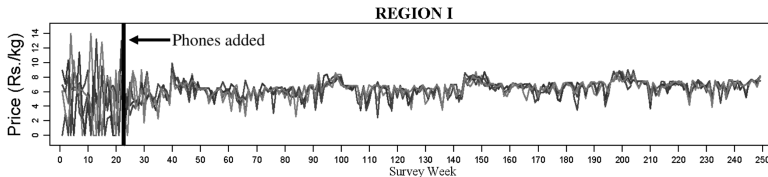


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(a) A price of zero is a boat that reached a beach where the buyers had gone. The catch was **dumped**. Today's fish cannot wait, cannot travel, cannot be stored.

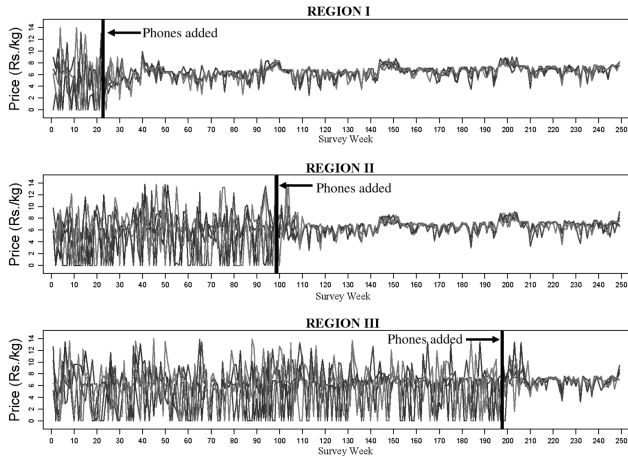
# The graph everyone remembers



(a) A price of zero is a boat that reached a beach where the buyers had gone. The catch was **dumped**. Today's fish cannot wait, cannot travel, cannot be stored.

(b) The average hardly moves — the **spread** collapses. Fishermen earn 8% more, consumers pay 4% less, and the 5-8% of the catch that used to rot is now sold. Nobody caught an extra fish.

# Three regions, three dates

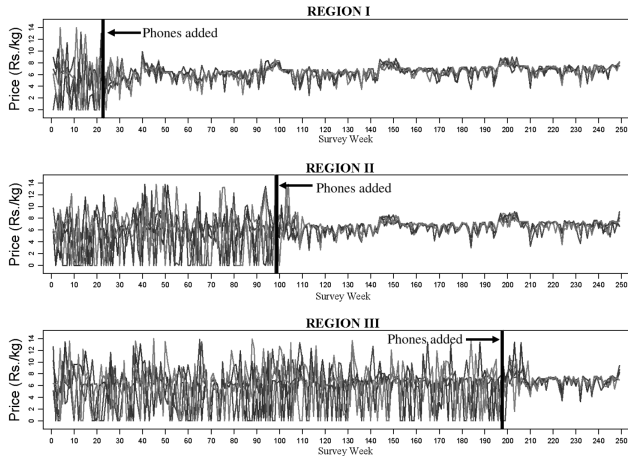


Phones reached region I at week 23, region II at week 98, region III at week 198.

Each series breaks at **its own** date, not at a common one.

Evidence, not coincidence: the same change, three times, whenever the phones arrived.

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Evidence, not coincidence: the same change, three times, whenever the phones arrived.

A natural experiment: theory first, then data that could have refuted it.

# What the phones did

Mobile service reached the coast **region by region**, 1997–2001. Fishermen called ahead and chose the beach.

|                      | Before | After          |
|----------------------|--------|----------------|
| Price dispersion     | 60–70% | $\leq 15\%$    |
| One-price violations | 50–60% | $\approx$ none |
| Catch wasted         | 5–8%   | <b>0</b>       |
| Fishermen's profits  | —      | +8%            |
| Consumer price       | —      | –4%            |

Jensen, *QJE* 122(3), 2007.

## Question

Profits up *and* prices down.  
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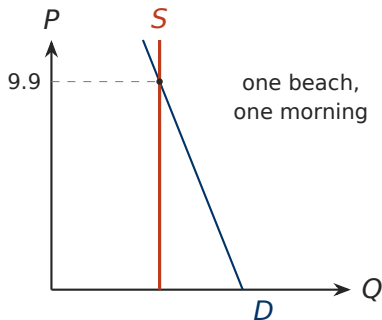
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Principle 6, measured: the signal must be **seen** to work.

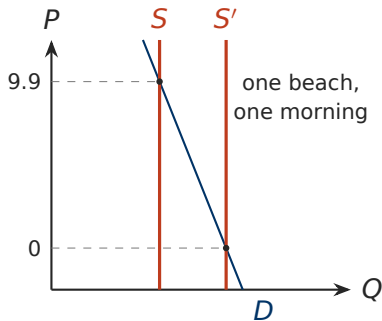
## Why were the price swings so violent?



Today's catch is **already caught**: supply for the morning is a vertical line.

Local buyers are few, the market shuts at 8:00, the fish will not keep. Demand at one beach is **steep**.

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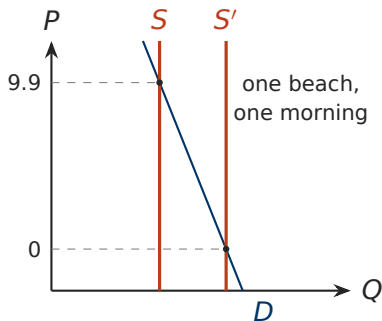


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## The chapter in one line

How far the price moves depends on how **steep** — how inelastic — demand is. Elasticity measures that.

# Your problem this semester

You build websites for local businesses.

You charge \$200 a site and sell 12 a month.

Your costs are rising — including the opportunity cost of your own time.

You are thinking about \$250.

## Question

The law of demand says you will sell fewer. Fine.

How many fewer?

And does your revenue go up or down?

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“Demand slopes down” does not answer either question.

You need a number.

# Price elasticity of demand

## Elasticity

A measure of how much quantity demanded or supplied **responds** to a change in one of its determinants

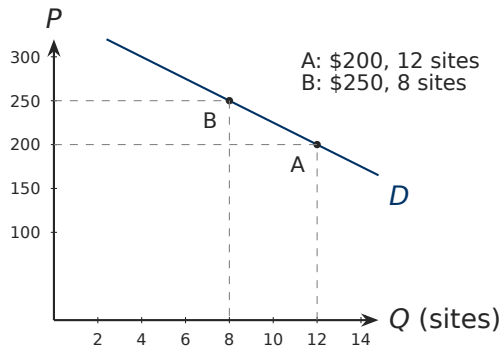
## Price elasticity of demand

$$E_d = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}$$

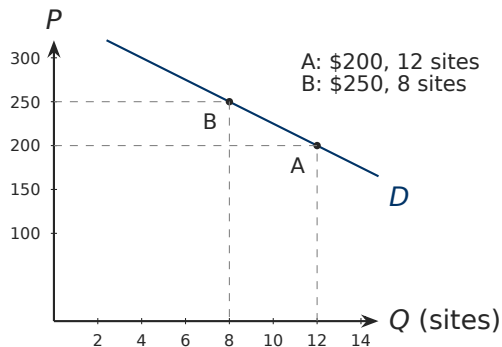
Along a demand curve  $P$  and  $Q$  move in **opposite** directions, so the ratio is negative. Convention: **drop the minus sign** and compare sizes.

*Percentages*, not units: rupees or dollars, kilos or tons, the number is the same.

## First try: the obvious calculation



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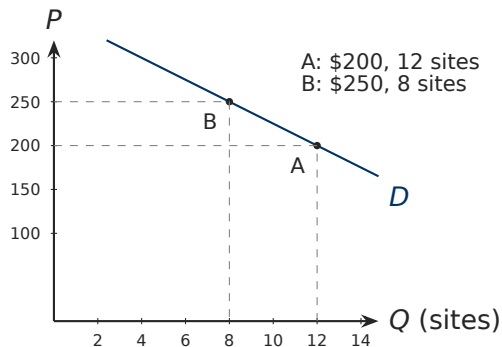
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$$\% \Delta P = (250 - 200) / 200 = 25\%$$

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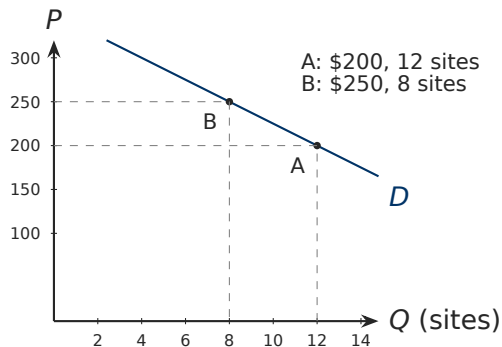
**B** → **A** (lower it again)

$$\% \Delta P = (200 - 250) / 250 = -20\%$$

$$\% \Delta Q = (12 - 8) / 8 = 50\%$$

$$E_d = 50 / 20 = 2.50$$

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Same two points, two answers. The base you divide by does the damage.

# The midpoint method

**Divide by the average, not by the starting point**

$$\% \Delta Q = \frac{Q_2 - Q_1}{(Q_1 + Q_2)/2} \quad \% \Delta P = \frac{P_2 - P_1}{(P_1 + P_2)/2} \quad E_d = \left| \frac{\% \Delta Q}{\% \Delta P} \right|$$

The midpoint is the number halfway between the two values. Going up or going down now gives the **same** elasticity.

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\$250, 12 → 8 sites:

$$\% \Delta Q = \frac{-4}{10} = -40\% \quad \% \Delta P = \frac{50}{225} = 22.2\% \quad E_d = \frac{40}{22.2} = 1.8$$

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Quantity moves **almost twice** as hard as price. Hold on to that 1.8.

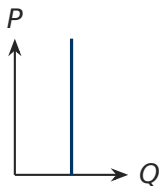
# The vocabulary

| Elasticity     | Name                | Meaning                                     |
|----------------|---------------------|---------------------------------------------|
| $E_d = 0$      | perfectly inelastic | quantity does not move at all               |
| $E_d < 1$      | inelastic           | quantity moves <i>less</i> than price, in % |
| $E_d = 1$      | unit elastic        | they move together                          |
| $E_d > 1$      | elastic             | quantity moves <i>more</i> than price, in % |
| $E_d = \infty$ | perfectly elastic   | any price change wipes out the quantity     |

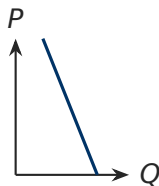
The dividing line is **1**: it is the line that decides what happens to revenue.

Same axes, same scale: the **flatter** the demand curve, the more elastic.

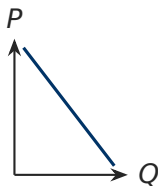
# The five cases



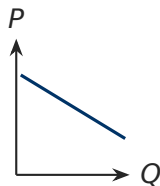
$E_d = 0$   
perfectly  
inelastic



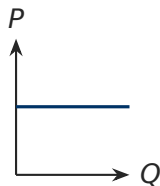
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**Vertical:** insulin for a diabetic this week · the sardines already in the boat

**Horizontal:** one farmer's wheat · one seller in a perfectly competitive market — charge a cent more and sell nothing

# What makes demand elastic

| Determinant              | Elastic when                | Example                         |
|--------------------------|-----------------------------|---------------------------------|
| Close substitutes        | they exist                  | butter vs. margarine            |
| Necessity or luxury      | it is a luxury              | sailboats, not insulin          |
| Definition of the market | the market is <b>narrow</b> | Honda Civics vs. cars           |
| Time horizon             | the horizon is long         | gasoline: a week vs. five years |

**Narrow beats broad.** “Food” has no substitute; “vanilla ice cream” has forty.

**Time is the quiet one.** Gas doubles: this week you still drive to work; over five years you change car, job, home.

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A **brand** is a narrow market — which is why one seller’s demand is elastic when the industry’s is not.

## Rank them

### Question

Which has the more elastic demand, and why?

- (a) Insulin or vanilla ice cream
- (b) Gasoline over one week or gasoline over five years
- (c) Cigarettes or Marlboros
- (d) Fish at one beach at 7:45 AM or fish in Kerala over a month

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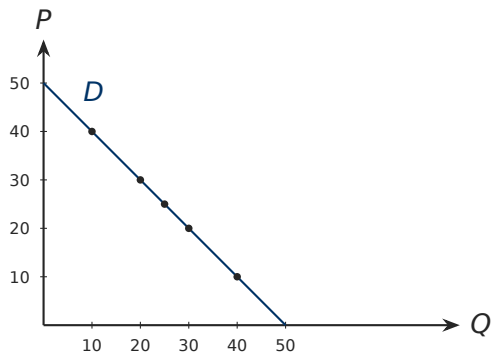
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- (d) The month. At 8:00 the unsold fish is worth nothing — hence 0 to 9.9.

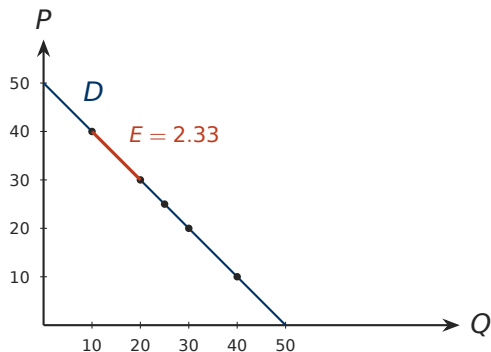
# Elasticity is not the slope



The Ch 4 curve,  $Q_d = 50 - P$ . One straight line, **constant slope**.

| $P$                 | $Q$                 | $\% \Delta Q / \% \Delta P$ | $E_d$ |
|---------------------|---------------------|-----------------------------|-------|
| 40 $\rightarrow$ 30 | 10 $\rightarrow$ 20 | 66.7/28.6                   |       |
| 30 $\rightarrow$ 20 | 20 $\rightarrow$ 30 | 40/40                       |       |
| 20 $\rightarrow$ 10 | 30 $\rightarrow$ 40 | 28.6/66.7                   |       |

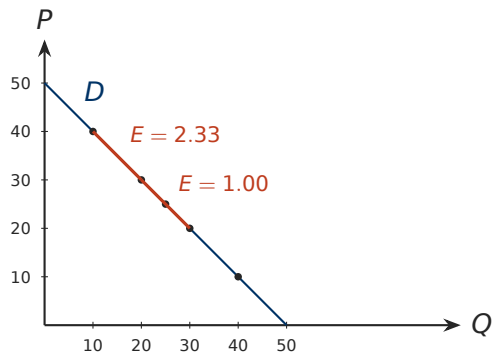
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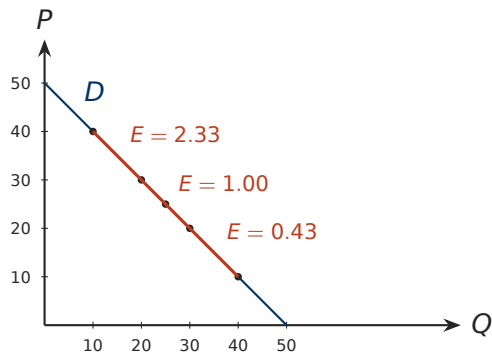
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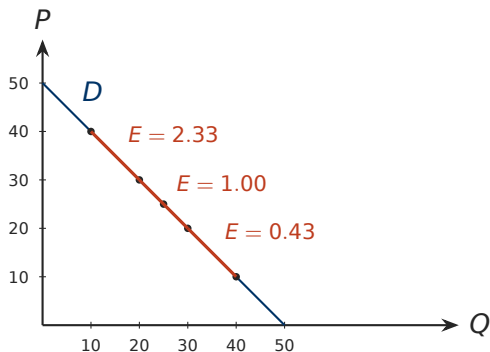
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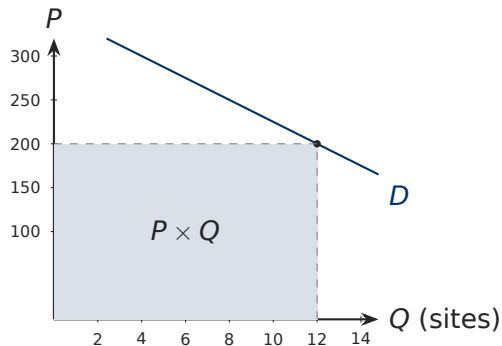
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**High price:** elastic. **Low price:** inelastic.

Elasticity **falls** as you move down the curve. Never say “the” elasticity of a linear curve.

# Total revenue



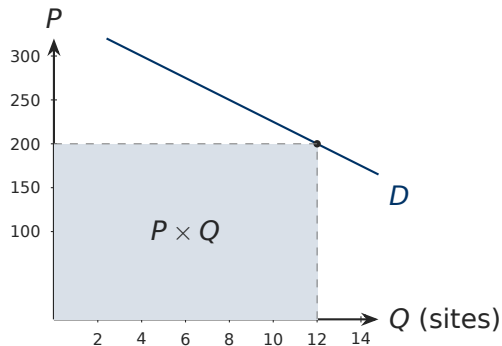
## Total revenue

$TR = P \times Q$  — what sellers take in, what buyers spend. The **area** of the rectangle under the point.

Raise the price and two things happen:

- + every unit you still sell earns more
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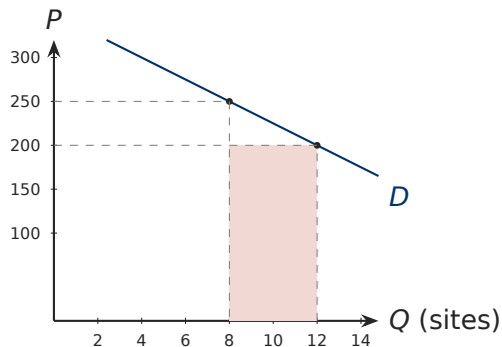
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The rectangle gets taller and narrower. Whether the **area** grows depends on which change is bigger in percent — on  $E_d$ .

## Elastic: the price increase backfires

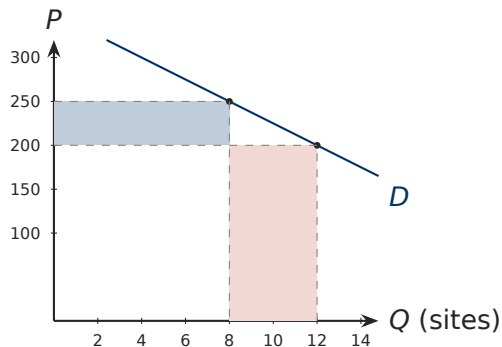


lost: fewer sites · gained: higher price

$E_d = 1.8$ , elastic.

|        | $P$   | $Q$ | $TR$           |
|--------|-------|-----|----------------|
| Before | \$200 | 12  | \$2,400        |
| After  | \$250 | 8   | <b>\$2,000</b> |

## Elastic: the price increase backfires

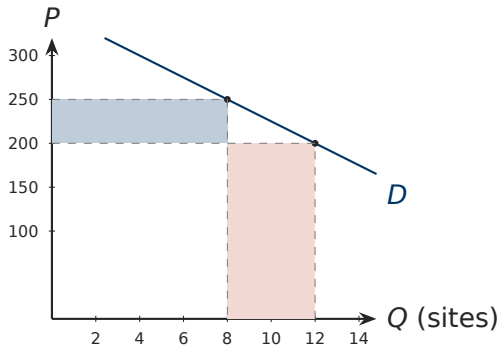


lost: fewer sites · gained: higher price

$E_d = 1.8$ , elastic.

|        | $P$   | $Q$ | $TR$           |
|--------|-------|-----|----------------|
| Before | \$200 | 12  | \$2,400        |
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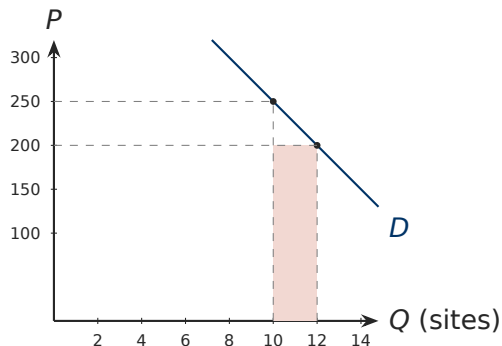
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Quantity fell 40%, price rose 22%.  
The loss is bigger.

## When demand is elastic

Price up  $\Rightarrow$  revenue **down**. Price  
down  $\Rightarrow$  revenue **up**.

## Inelastic: the same increase pays

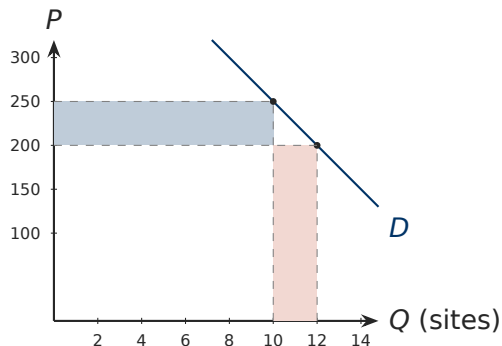


a steeper curve:  $12 \rightarrow 10$ , not  $12 \rightarrow 8$

Same price rise, but now  $E_d = 0.82$ .

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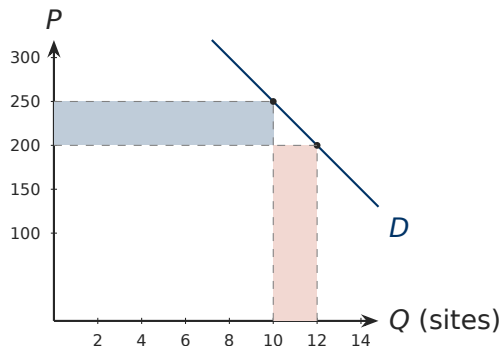


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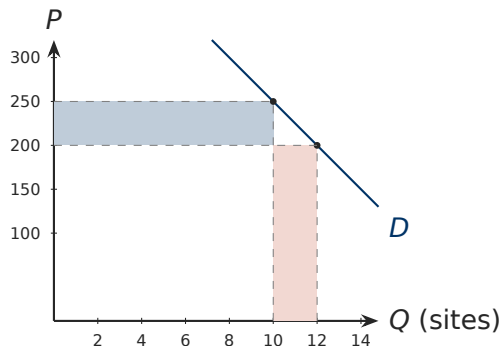
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Quantity fell 18%, price rose 22%.  
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Price up  $\Rightarrow$  revenue **up**. Price down  
 $\Rightarrow$  revenue **down**.

## Inelastic: the same increase pays



a steeper curve:  $12 \rightarrow 10$ , not  $12 \rightarrow 8$

Nothing about the good changed.  
Only the **number**.

Same price rise, but now  $E_d = 0.82$ .

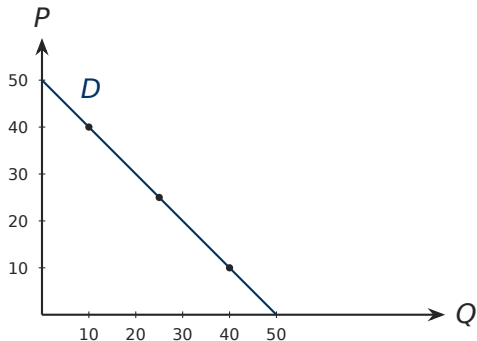
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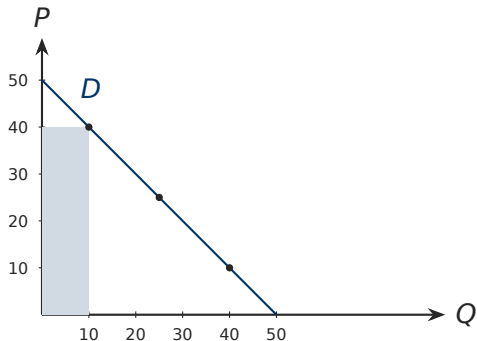
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## Revenue along the Ch 4 curve



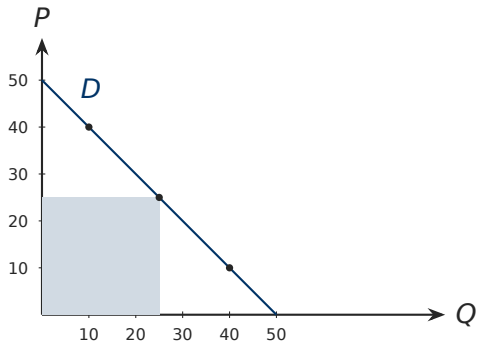
| $P$ | $Q$ | $TR$ |
|-----|-----|------|
| 40  | 10  | 400  |
| 30  | 20  | 600  |
| 25  | 25  | 625  |
| 20  | 30  | 600  |
| 10  | 40  | 400  |

# Revenue along the Ch 4 curve



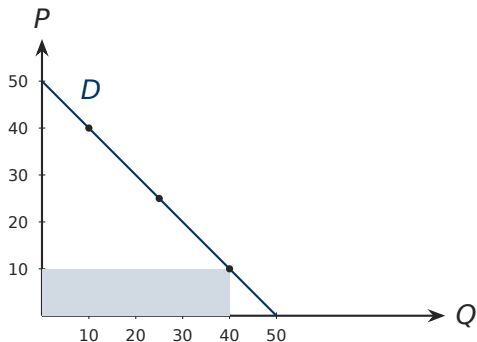
| $P$ | $Q$ | $TR$ |              |
|-----|-----|------|--------------|
| 40  | 10  | 400  | elastic here |
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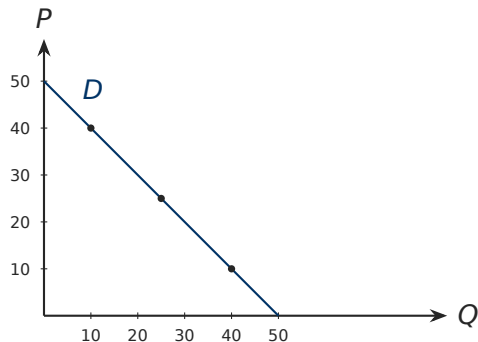
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Revenue peaks where  $E_d = 1$ , at the midpoint.

Above it, a price cut raises revenue;  
below it, a price cut lowers revenue.

### Question

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One rule, two different numbers, opposite answers.

## Two more demand elasticities

### Income elasticity

$$E_I = \frac{\% \Delta Q_d}{\% \Delta \text{income}}$$

**Positive:** normal good

**Negative:** inferior good (bus rides, instant noodles)

Among normal goods: necessities have small positive values (food, fuel), luxuries large ones (flights, jewelry)

Keep the sign here. It carries the information.

### Cross-price elasticity

$$E_{xy} = \frac{\% \Delta Q_d \text{ of good } x}{\% \Delta P \text{ of good } y}$$

**Positive:** substitutes — Pepsi's price up, Coke's sales up

**Negative:** complements — printers dearer, ink sales down

Near zero: unrelated goods

This is the Ch 4 shifter table, now with a number attached.

# Price elasticity of supply

## Price elasticity of supply

$$E_s = \frac{\% \Delta Q_s}{\% \Delta P}$$

How much sellers respond to a better price. Same arithmetic, same midpoint method.

### Inelastic supply

Beachfront land · a Rembrandt ·  
this morning's catch, already  
landed

No price makes more of it appear

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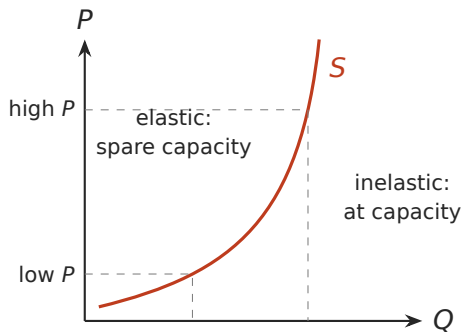
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**Time again.** Over a day, output is nearly fixed; over years, firms enter and build. **Supply is far more elastic in the long run.**

# Supply near capacity

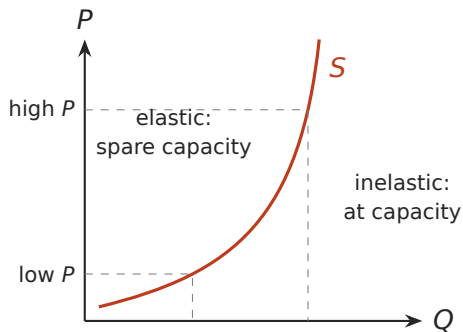


Supply elasticity is not one number either.

**Low output:** idle machines, unused hours. A small price rise brings out a lot more. **Elastic**.

**Near capacity:** plants running flat out. Even a large price rise brings out little. **Inelastic**.

# Supply near capacity



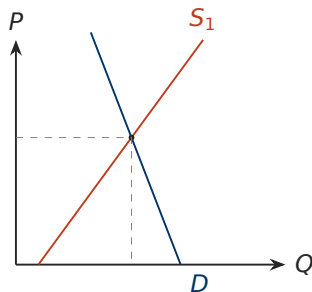
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Same logic as the Kerala beach: when quantity cannot adjust, all the adjustment lands on the **price**.

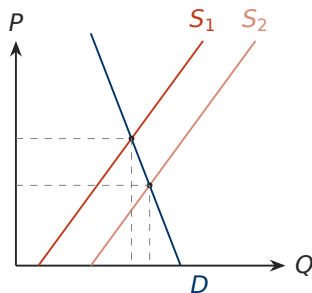
## Application 1: good news for farming, bad news for farmers



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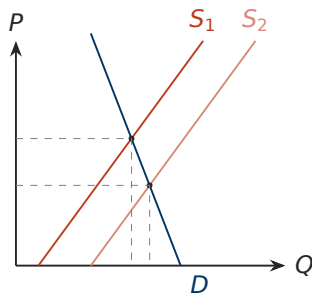


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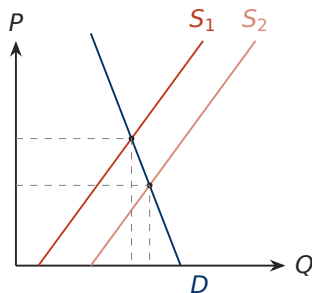
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Each farmer is rational; together they cut their incomes.

U.S. farm employment: 41% of the workforce in 1900, 1.9% in 2000.

## Application 2: why OPEC could not hold the price up

### Short run

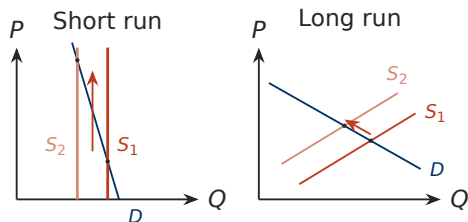
Supply (wells already drilled) and demand (your car, your furnace) are both **inelastic**

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### Long run

New fields open, cars get efficient — **both curves turn elastic**

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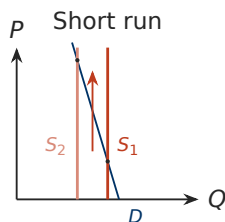


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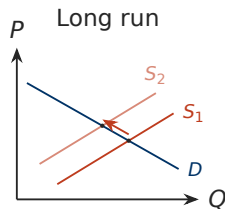
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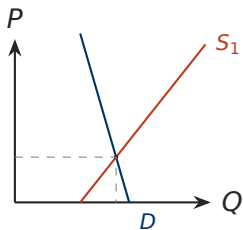
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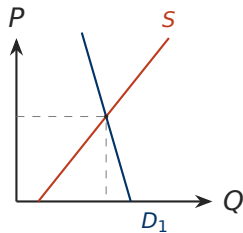
Crude, nominal: about \$3 a barrel before Oct 1973, \$11.65 posted in Jan 1974, over \$35 in 1980 — then a slide from \$26 to near \$10 during 1986.

## Application 3: two ways to fight drug use

### 1. Interdiction

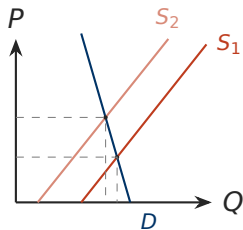


### 2. Treatment and education



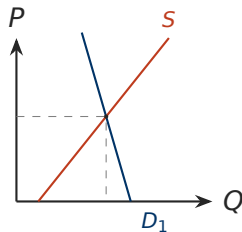
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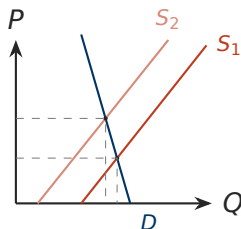
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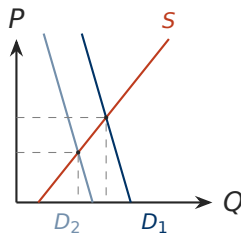
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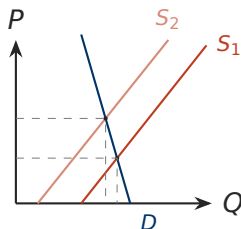
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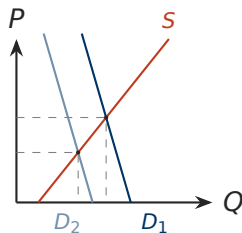
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Both cut the quantity consumed. Only one cuts the **sellers' revenue** — what finances the trade, and what users steal to pay.

The argument turns on demand being inelastic. If it were elastic, interdiction would cut seller revenue too.

## Exam-style: cigarettes

### Question

The price elasticity of demand for cigarettes is about 0.2. A pack costs \$5.

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- (b) Is such a tax more effective in the short run or the long run?
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(c) More sensitive: less income, and not yet addicted. Addiction is what makes adult demand inelastic.

# Summary

## Measuring

- Elasticity = % change in quantity over % change in price; report it positive
- Midpoint method: divide by the average, so up and down agree
- Elastic  $> 1$ , inelastic  $< 1$ , unit = 1
- Elastic when there are substitutes, when it is a luxury, when the market is narrow, when there is time
- A linear demand curve has every elasticity along it

## Using

- $TR = P \times Q$ ; elastic  $\Rightarrow$  price and revenue move *opposite*, inelastic  $\Rightarrow$  *together*
- Revenue peaks where  $E_d = 1$
- Income elasticity signs normal vs. inferior; cross-price signs substitutes vs. complements
- Supply: inelastic when quantity is fixed or capacity is full, elastic in the long run
- Whichever side cannot adjust, the **price** does the adjusting

**Next: Ch 6, Supply, Demand, and Government Policies.**

**MindTap: Ch 5 homework due Sunday, 11 PM.**